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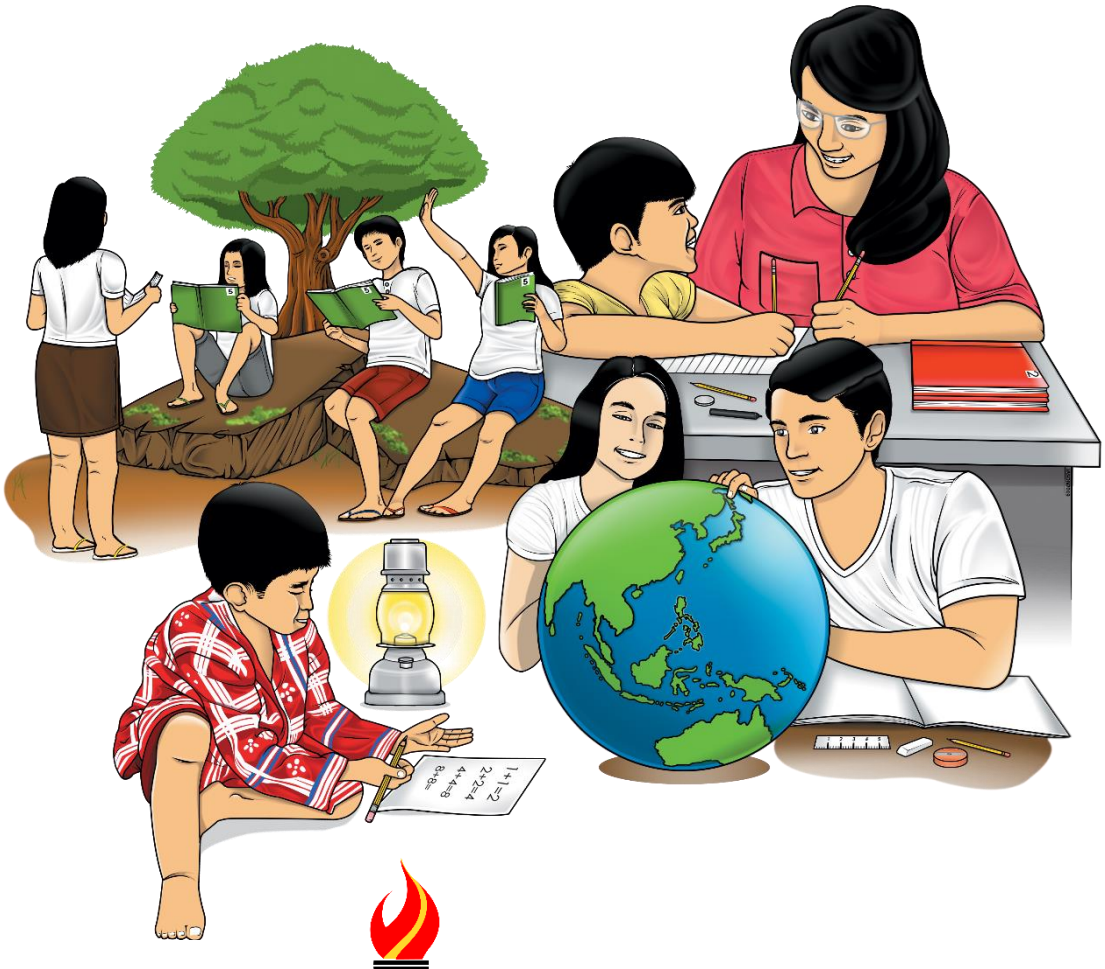
Senior High School

Pre-Calculus
Quarter 1



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What I Know?

Choose the letter of the best answer.

____**1**. What do you call to the locus of a point which moves at a constant distance from a fixed point called its *center*?

A. Circle B. Ellipse C. Hyperbola D. Parabola

____**2**. It is the intersection of a plane and a double-napped cone.

A. circle B. parabola C. ellipse D. conic

____**3** Which of the following is the standard form of the circle?

A. $(x - h)^2 + (y - k)^2 = r^2$

B. $(x - h)^2 = 4c(y - k)$

C. $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

D. $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

____**4.** In the standard form of equation of a circle, $(x - h)^2 + (y - k)^2 = r^2$, (h, k) is known as _____.

A. center B. vertex C. directrix D. focus

For nos. 5-6. Given: Point (1, 4) is on a circle whose center is at (-2, -3).

____**5.** What is the radius of the circle?

A. $\sqrt{58}$ B. 58 C. 7.61 D. $2\sqrt{39}$

____**6.** What is the standard form of the equation of the circle?

A. $(x - 2)^2 + (y - 3)^2 = 58$

B. $(x + 2)^2 + (y + 3)^2 = \sqrt{58}$

C. $(x - 3)^2 - (y - 2)^2 = \sqrt{58}$

D. $(x + 2)^2 + (y + 3)^2 = 58$

____**7.** Find the standard form of a circle having a center at (0,0) and passing through the point (3, -4)

A. $x^2 + y^2 = 1$

B. $x^2 + y^2 = 9$

C. $x^2 + y^2 = 7$

D. $x^2 + y^2 = 25$

____**8.** What is the radius of the equation in number 18?

A. 1 B. 2 C. 4 D. 5

____**9.** What is the center of this equation of a circle $(x - 3)^2 + y^2 = 16$?

A. (0, 3) B. (3,0) C. (0,-3) D. (-3,0)

____**10.** Find the radius of the equation given in number 9.

A. 1 B. 3 C. 4 D. 16

Lesson 1 Circles

What I Need to Know?

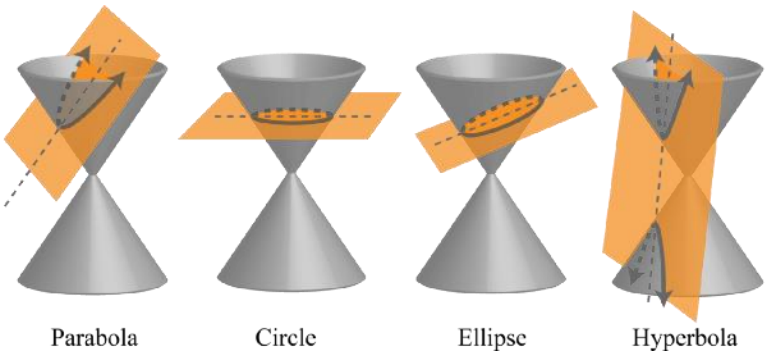
Expected Learning Outcome/s: **(LC Codes:** STEM_PC11AG-Ia-1-3**)**

1. Illustrate the different types of conic sections.
2. Define a circle.
3. Graph a circle given an equation in center – radius form.

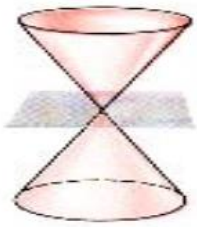
What's In?

The different types of **conic sections** are circle, parabola, ellipse, hyperbola, and degenerate cases. If a plane is made to cut a right circular cone, the section obtained is called a *conic section*, or simply a *conic*.

The shape of the conic will depend on the position of the cutting plane. If the cutting plane is parallel to the base of the cone, we have a *circle*. If the cutting plane intersects only one cone to form an unbounded curve, the section is called *parabola*. If the cutting plane intersects only one cone to form a bounded curve, the section is called an *ellipse* and if the cutting plane intersects both nappes or sheets, the conic section is called a hyperbola (Figure 1.1).

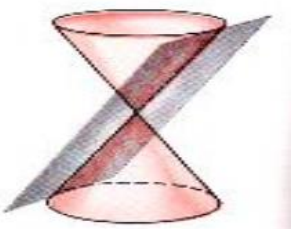


Other exceptional types of conic sections called *degenerate conics* are the point, two coincident lines (one line) and two intersecting lines (two lines). These are obtained when the cutting plane passes through the vertex (intersection of two cones). See Figures 1.2, 1.3 and 1.4.



Point
Figure 1.2

Intersecting lines
Figure 1.3



Line
Figure 1.4

What's New?

Circle - is the locus of a point which moves at a constant distance from a fixed point called its *center*. The constant distance of any point from the center is called the *radius*.

Let P(x,y) be any point on the circle, C(h,k) are the center and r > 0 is the radius (Figure 1.5).

By definition,

$$PC = r$$
$$\sqrt{(x-h)^2 + (y-k)^2} = r$$
$$(x-h)^2 + (y-k)^2 = r^2$$

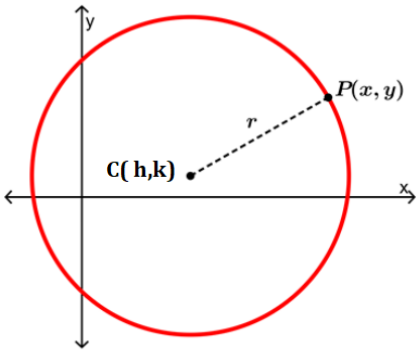


Figure 1.5

This is the standard equation of a circle of radius r whose center is at C(h,k).

If the center is at the origin, then h = k = 0 and the equation of the circle becomes **$x^2 + y^2 = r^2$.**

What is It?

Standard form of the circle

An equation of a circle centered at (h,k) and radius is r >0 is:	**An equation of a circle centered at the origin and radius is r >0 is:**
$(x-h)^2+(y-k)^2 = r^2$	$x^2 + y^2 = r^2$

General form of the circle

The general equation of the circle may be taken in this form:

$$Ax^2 + Ay^2 + Cx + Dy + E = 0$$

or

$$x^2 + y^2 + Cx + Dy + E = 0$$

What's More?

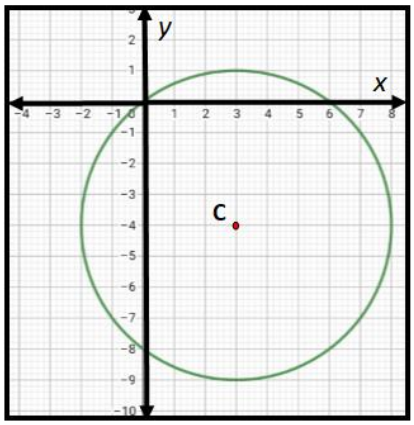
Finding the Equation of a Circle

Give the standard form of the circle satisfying the following given conditions.	
1. Center at the origin, radius 3 units a. Find the center b. Find the radius c. Find the equation of the circle	***Reasons*** a. C(0,0) center at the origin b. r = 3 units given c. $x^2 + y^2 = r^2$ formula $x^2 + y^2 = 3^2$ substitution $\boxed{\mathbf{x^2 + y^2 = 9}}$ simplify
2. Center (-3,6), radius 6 units a. Find the center b. Find the radius c. Find the equation of the circle	a. C(-3,6) given b. r = 6 units given c. $(x - h)^2+(y - k)^2 = r^2$ formula $[x-(-3)]^2+(y - 6)^2 = 6^2$ substitution $\boxed{\mathbf{(x+3)^2 + (y-6)^2 = 36}}$ simplify
3. Center (4,5), tangent to y-axis a. Find the center b. Find the radius c. Find the equation of the circle	a. C(4,5) given b. r = 4 units using the distance formula $D=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$ Distance from A(4,5) to B(0,5) c. $(x-h)^2+(y-k)^2 = r^2$ formula $(x-4)^2+(y-5)^2 = 4^2$ substitution $\boxed{\mathbf{(x-4)^2+(y-5)^2 = 16}}$ simplify
4. Has a diameter with endpoints A(4,4) and B(-2,-4) a. Find the center b. Find the radius c. Find the equation of the circle	a. C(1,0) using Midpoint formula $=\frac{(x_1+x_2)}{2},\frac{(y_1+y_2)}{2}$ b. r = 5 units using the distance formula $\frac{1}{2}\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$ (half of the diameter) c. $(x-h)^2+(y-k)^2 = r^2$ formula $(x-1)^2+(y-0)^2 = 5^2$ substitution $\boxed{\mathbf{(x-1)^2+ y^2 = 25}}$ simplify

What I Have Learned?

- **Circle -** is the locus of a point which moves at a constant distance from a fixed point called its *center.*
- The constant distance of any point from the center is called the ***radius***.
- The equation of the circle centered at the origin and r >0 is $\mathbf{x^2 + y^2 = r^2}$.
- The equation of a circle centered at (h,k) and radius is r >0 is $\mathbf{(x-h)^2+(y-k)^2 = r^2}$

What I Can Do?

Give the standard form of the circle satisfying the following given conditions.	
1. Center at the origin, radius 7 units a. Find the center b. Find the radius c. Find the equation of the circle	a. _____ b. _____ c. _____
2. Center (2,-3), radius 4 units a. Find the center b. Find the radius c. Find the equation of the circle	a. _____ b. _____ c. _____ (Hint: Use distance formula in finding the radius)
3. Center (- 3,- 4), tangent to x-axis a. Find the center b. Find the radius c. Find the equation of the circle	a. _____ b. _____ c. _____
4. Has a diameter with endpoints A(-1,2) and B(5,6) a. Find the center b. Find the radius c. Find the equation of the circle	(Hint: Use midpoint formula in finding the center of the circle and distance formula in finding the length of the diameter. Radius is $\frac{1}{2}\lvert AB\rvert$.) a. _____ b. _____ c. _____
5. Given the illustration below 	a. _____ b. _____ c. _____

Short Answer. Supply the correct answer.

Assessment

_____1. These are the "curves" of intersection we get when the vertex is included.

_____2. It is a locus of all points in the plane having the same fixed positive distance, called the radius, from a fixed point, called the center.

_____3. It is the formula used in finding the center of a circle given the endpoints of a diameter.

_____4. It is the center of a circle whose equation is $(x-h)^2+(y-k)^2 = r^2$,

where $r > 0$.

_____5. What non- degenerate conic is formed when the cutting plane is parallel to the base of the cone?

_____6. Given the equation of the circle $(x - 7)^2+(y+3)^2 - 81 = 0$, what is the measure of the radius r ?

_____7. If (- 5,6) is the center of circle *O* and tangent to y – axis, what is the equation of the circle?

_____8. YES OR NO: Is point (1,5) lies on a circle whose center is (-1,5) and passing through point (-1,7)?

_____9. What is the standard form of the equation of the circle whose origin is the center and 6 units is the radius?

_____10. Convert $x^2 + y^2 - 6x + 4y - 12 = 0$ in standard form.

Additional Activities

General Directions: Avoid erasure. Box your final answer.

I. Find the center and radius of the given circle.

Equation of the circle	*Center*	*Radius*
$x^2 + y^2 = 81$		
$(x +5)^2 + y^2 = 16$		
$\frac{(x-2)^2}{9} + \frac{(y+1)^2}{9} = 1$		
$x^2 + (y - 3)^2 = \frac{4}{25}$		

II. Write the standard form of the circle with the given properties.

Given	Standard form
C(1,-2), radius 6 units	
C(-2,-5), radius $\frac{3}{4}$ units	
Center at the origin and (2,-3) is a point on the circle.	
Endpoints of a diameter of a circle are (2,6) and (5,-2)	

What I Know?

Choose the letter of the best answer.

____**1**. What do you call a curve that has points which are equidistant from the fixed point and the given line?

A. Circle B. Ellipse C. Hyperbola D. Parabola

____**2**. This is the length of space from the focus to the vertex. It is also the distance between the directrix line and the vertex.

A. Latus Rectum
B. Directrix line
C. Focal Distance
D. Axis of Symmetry

____**3**. Which of the following is the vertex form of parabola that opens to the left?

A. $(y - k)^2 = -4c(x - h)$
B. $(y - k)^2 = 4c(x - h)$
C. $(x - h)^2 = -4c(y - k)$
D $(x - h)^2 = 4c(y - k)$

For numbers 4 - 10. Given the equation of parabola : $x^2 - 4x + 4y - 8 = 0$.

____**4.** What is the standard equation of the parabola?

A. $(x - 2)^2 = -4(y - 3)$
B. $(x + 2)^2 = -4(y + 3)$
C. $(x - 2)^2 = 4(y - 3)$
D. $(x + 2)^2 = 4(y + 3)$

____**5.** What is the graph of this parabola?

A. opens upward
B. opens to the right
C. opens downward
D. opens to the left

____**6.** Solve the measure of the focal distance.

A. $\frac{1}{2}$ B. $\frac{3}{4}$ C. 1 D. 8

____**7.** Find the length of latus rectum?

A. 2 B. 4 C. 6 D. 8

____**8.** What is the coordinate of the vertex?

A. (2, 3) B. (-2, 3) C. (-2, -3) D. (2, -3)

____**9.** What is the coordinate of the focus?

A. (2, 2) B. (-2, 2) C. (-2, -2) D. (2, -2)

____**10.** Which are the endpoints of the latus rectum?

A. (2,0) , (2, 4) B. (0,2), (4,2) C. (-2,0) , (-2, 4) D. (0,-2), (4,-2)

Lesson 2 Parabolas

What I Need to Know?

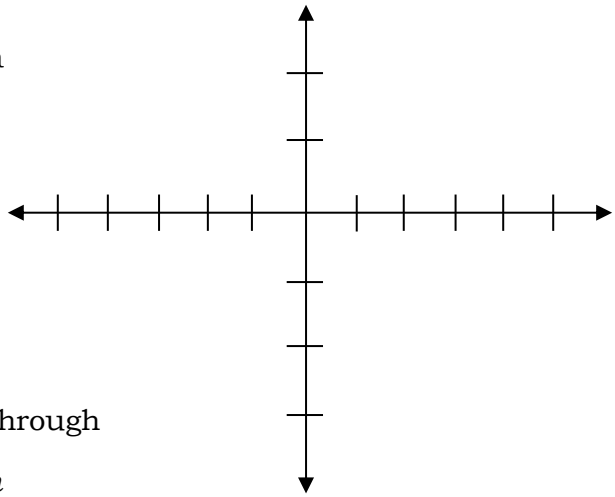
Expected Learning Outcomes:

(LC Codes: STEM_PC11AG-Ia-5, STEM_PC11AG-Ib-1**)**

1. Define a parabola.
2. Determine the standard form of equation of a parabola.

What's In?

"FOLLOWING DIRECTIONS"

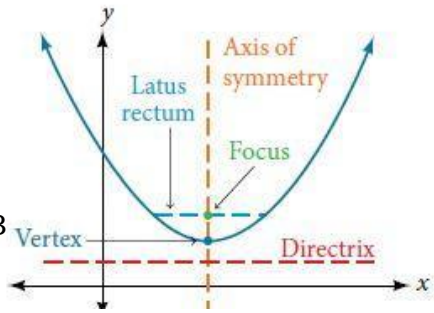


1. Graph the following ordered pairs in a rectangular coordinate plane:
   A(0, 0), B(0, 2), C(4, 2), D(-4,2)
2. Draw a curve that passes through the points of A, C, and D
3. Draw a line that passes through the points of A and B
4. Draw a horizontal line which passes through the equation $y = -2$. Name it as *line m*
5. Draw a line segment connecting the points of B and C.
6. From point C, draw a line segment perpendicular to a line in number 3. Name the line segment as l.

Guide Questions:

1. Describe the graph drawn in number 2?
2. What is the distance of point A from point B?
3. Describe the distance from point B to C and the distance from point B to D.
4. What is the length of line segment BC($\underline{BC}$)?
5. What is the length of line segment l?
6. Describe the length of $\underline{BC}$ and line segment l?

What's New?



From the activity above, Let B be a given point, and *line m*, a given line not containing B. The set of all points P such that its distances from B

and from *line m* are the same, is called a parabola.
The point B is its focus and the *line m* its directrix.

PARABOLA is defined as a curve that has points which are equidistance from the fixed point and the given line called directrix.

What is It?

To understand easily the concepts of this conic section, let us be familiarized with its parts and their definition.

PARTS OF A PARABOLA

1. **Vertex (V)** - It is the main point of the parabola which lies at the middle portion of the curve. It is also the midpoint of the fixed point of the curve and the given line.

2. **Focus (F)** - It is another important point of the parabola that is located inside of the curve. The focus is also known as the fixed point of the parabola.

3. **Focal Distance (c)-** This is the length of space from the focus to the vertex. It is also the distance between the directrix line and the vertex.

4. **Directrix line (D.L)** - It is a line which is outside the parabola and perpendicular to the axis of symmetry.

5. **Latus Rectum(LR)-** This is a chord that is parallel to the directrix line and intersecting the parabola at the point of focus. The end points of the latus rectum are defined as LR_1 and LR_2. The length of the latus rectum is always twice of the focal distance. The half of the latus rectum is called the semi-latus rectum.

6. **Axis of Parabola -** It is also known as axis of symmetry because it is the line that divides the parabola into two equal parts. The axis of the parabola passes through the points of Vertex and Focus; and this line is perpendicular to the directrix and the latus rectum.

What's More?

Find the equation of the parabola in vertex form using the indicated conditions and sketch the graph.	
Steps	**Solution**
1. Vertex (0,0) and Focus (-2,0) a. List down all the given b. Identify the graph of parabola c. Substitute the needed values to the equation of parabola d. simplify	a. $V(0,0)$ $F(-2,0)$ $c = 2$ $2c = 4$ $4c = 8$ b. $(y-h)^2 = -4c(x-h)$ → opens left c. $(y-0)^2 = -8(x-0)$ **d.** $y^2 = -8x$

2. Directrix line equation $y - 6 = 0$, vertex at (0,5) a. List down all the given b. Identify the graph of parabola c. Substitute the needed values to the equation of parabola d. simplify	a. $V(0,5)$ $DL: y = 6$ $c = 1$ $2c = 2$ $4c = 4$ b. $(xh)^2 = -4c(y-k) \rightarrow$ opens down c. $(x-0)^2 = -4(y-5)$ **d.** $x^2 = -4(y-5)$
3. Vertex (1, -4) and $LR_1(2,-2)$ a. List down all the given b. Identify the graph of parabola c. Substitute the needed values to the equation of parabola d. simplify	a. $V(1,-4)$ $LR_1(2,-2)$ $c = 1$ $2c = 2$ $4c = 4$ b. $(y-h)^2 = 4c(x-h) \rightarrow$ opens right c. $(y-(-4))^2 = 4(x-1)$ **d.** $(y+4)^2 = 4(x-1)$

What I Have Learned?

To graph a parabola in a rectangular coordinate system given its standard form equation,

1. Determine which of the standard forms applies to the given equation:
$(y-h)^2 = \pm 4c(x-h)$ or $(x-h)^2 = \pm 4c(y-k)$

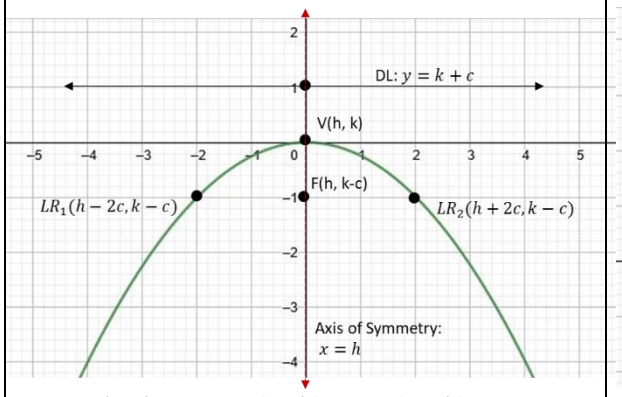
2. Use the standard form identified in Step 1 to determine the axis of symmetry, focus, equation of the directrix, and endpoints of the latus rectum.

a. If the equation is in the form $(y-h)^2 = \pm 4c(x-h)$, then
- the axis of symmetry is $y = k$
- set $4c$ equal to the coefficient of x in the given equation to solve for c. If $c > 0$, the parabola opens right. If $c < 0$, the parabola opens left.
- use c to find the coordinates of the focus, $(h \pm c, k)$
- use c to find the equation of the directrix, $x = h \pm c$
- use c to find the endpoints of the latus rectum. For a graph that opens left,use LR_1 (h - c, k - 2c) and LR_2 (h - c, k + 2c). For a graph that opens right, LR_1 (h +c, k - 2c) and LR_2 (h +c, k + 2c)

b. If the equation is in the form $(x-h)^2 = \pm 4c(y-k)$, then
- the axis of symmetry is $x = h$
- set $4c$ equal to the coefficient of y in the given equation to solve for c. If $c > 0$, the parabola opens up. If $c < 0$, the parabola opens down.
- use c to find the coordinates of the focus, $(h, k \pm c)$
- use c to find equation of the directrix, $y = k \pm c$
- use c to find the endpoints of the latus rectum. For a graph that opens down, use LR_1 (h - 2c, k - c) and LR_2 (h + 2c, k - c). For a graph that opens up, LR_1 (h - 2c, k + c) and LR_2 (h + 2c, k + c)

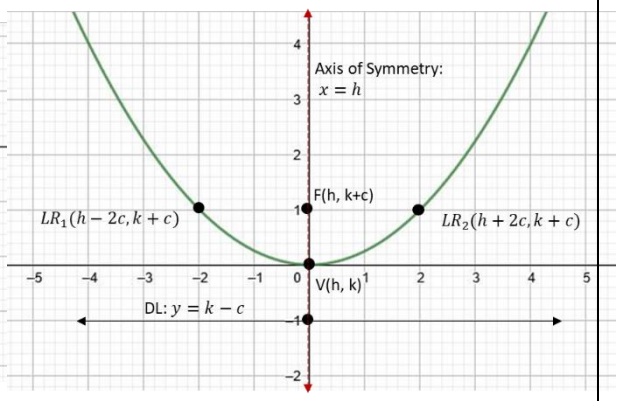
3. Plot the focus, directrix, and latus rectum, and draw a smooth curve to form the parabola.

OPENS DOWN



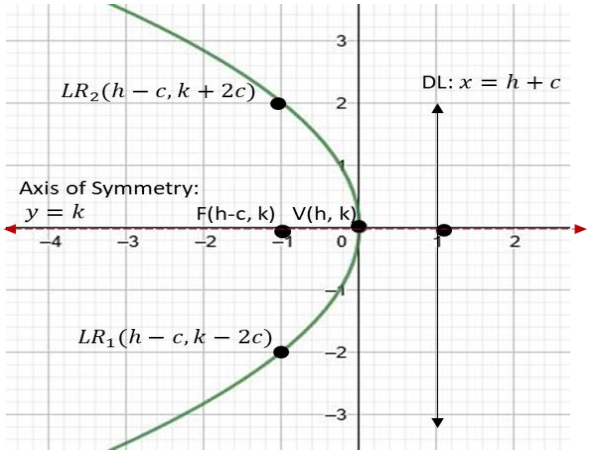
- Standard Form: $(x - h)^2 = -4c(y - k)$
- Vertex: V (h, k)
- Focus: F (h, k-c)
- Directrix Line: y = k + c
- Endpoints of LR:
  LR_1 (h-2c, k-c)
  LR_2 (h+2c, k-c)
- Axis of Symmetry: x = h

OPENS UP



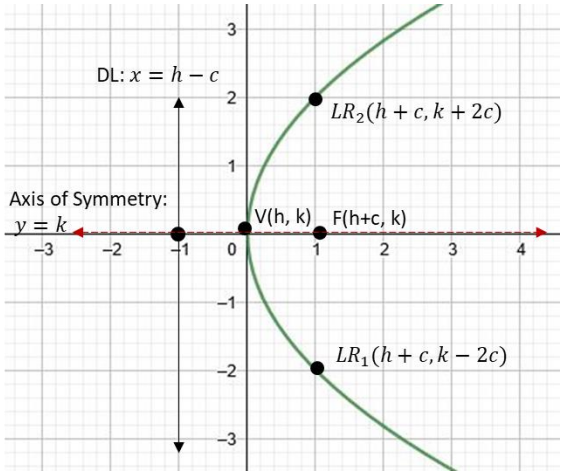
- Standard Form: $(x - h)^2 = 4c(y - k)$
- Vertex: V (h, k)
- Focus: F (h, k+c)
- Directrix Line: y = k - c
- Endpoints of LR:
  LR_1 (h-2c, k+c)
  LR_2 (h+2c, k+c)
- Axis of Symmetry: x = h

OPENS LEFT



- Standard Form: $(y - k)^2 = -4c(x - h)$
- Vertex: V (h, k)
- Focus: F (h - c, k)
- Directrix Line: x = h + c
- Endpoints of LR:
  LR_1 (h - c, k - 2c)
  LR_2(h - c, k + 2c)
- Axis of Symmetry: y = k

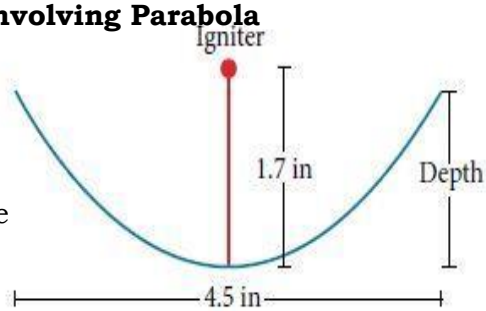
OPENS RIGHT



- Standard Form: $(y - k)^2 = 4c(x - h)$
- Vertex: V (h, k)
- Focus: F (h + c, k)
- Directrix Line: x = h - c
- Endpoints of LR:
  LR_1 (h +c, k - 2c)
  LR_2 (h +c, k + 2c)
- Axis of Symmetry: y = k

Solving Applied Problem Involving Parabola

A cross-section of a design for a travel-sized solar fire starter is shown in the figure. The sun's rays reflect off the parabolic mirror toward an object attached to the igniter. Because the igniter is located at the focus of the parabola, the reflected rays cause the object to burn in just seconds.



a. Find the equation of the parabola that models the fire starter. Assume tha vertex of the parabolic mirror is the origin of the coordinate plane.
b. Use the equation found in part (a) to find the depth of the fire starter.

Solution:

a. The vertex of the dish is the origin of the coordinate plane, so the parabola will take the standard form$(x-h)^2 = \pm 4c(y-k)$, where $c > 0$. The igniter, which is the focus, is 1.7 inches above the vertex of the dish. Thus we have $c = 1.7$.

$(x-h)^2 = 4c(y-k)$ → Standard form of upward-facing parabola with vertex (0, 0)

$(x-0)^2 = 4(1.7)(y-0)$ Substitute 1.7 for c.

$x^2 = 6.8y$ **equation that models the fire starter**

b. The dish extends $\frac{4.5}{2} = 2.25$ inches on either side of the origin. We can substitute 2.25 for x in the equation from part (a) to find the depth of the dish.

$x^2 = 6.8y$ → Equation found in part (a).

$(2.25)^2 = 6.8y$ Substitute 2.25 for x.

$y = 0.74$ Solve for y.

The dish is about 0.74 inches deep.

Assessment

Given: , find the following:

Focal Distance
Vertex
Line
Focus
Length of Lactus Rectum
5. End of points of LR
6. Equation of Directrix
7. Axis of the Parabola
8. Graph of Parabola

What I Know?

Choose the letter of the best answer.

____**1**. What conic section is formed by a locus of points whose sum of the distance from the two fixed points is constant?

A. Circle B. Ellipse C. Hyperbola D. Parabola

____**2**. What do you call the point (±a, 0) in the the equation of an ellipse

$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$?

A. center B. vertices C. directrix D. focus

____**3** Find the equation of the ellipse using the indicated conditions : C(0, 0), Vertices at (± 5, 0) and Foci (±3, 0).

A. $9x^2 + 25y^2 + 400 = 0$
B. $9^2 + 25y^2 - 400 = 0$
C. $16x^2 + 25y^2 - 400 = 0$
D. $16x^2 - 25y^2 - 400 = 0$

____**4.** Given the condition in number 3, where is the major axis located at?

A. at the origin
B. along the y-axis
C. along the x-axis
D. (5, -3)

____**5.** Find the standard equation of the given equation in 3.

A. $\frac{x^2}{25}+\frac{y^2}{16}=1$ B. $\frac{x^2}{25}-\frac{y^2}{16}=1$ C. $\frac{x^2}{16}+\frac{y^2}{25}=1$ D. $\frac{x^2}{16}-\frac{y^2}{25}=1$

For numbers 6-10. Given the equation of ellipse : $25x^2 + 9y^2 = 225$

____**6.** Find the standard equation.

A. $\frac{x^2}{9}+\frac{y^2}{25}=1$ B. $\frac{x^2}{9}-\frac{y^2}{25}=1$ C. $\frac{x^2}{25}+\frac{y^2}{9}=1$ D. $\frac{x^2}{25}-\frac{y^2}{9}=1$

____**7.** Find the coordinates of the vertices.

A. (0, 5), (0, -5) B. (5, 0),(-5, 0) C. (0, 4), (0, -4) D. (4, 0), (-4,0)

____**8.** Find the coordinates of the foci.

A. (0, 5), (0, -5) B. (5, 0),(-5, 0) C. (0, 4), (0, -4) D. (4, 0), (-4,0)

____**9.** Find the endpoints of the covertices.

A. (0, -3) (0, 3) B. (-3, 0) (3, 0) C. (0, -6) (0, 6) D. (-6, 0) (6, 0)

____**10.** Where is the major axis located?

A. at the origin
B. along the y-axis
C. along the x-axis
D. undefined

Lesson 3 Ellipse

What I Need to Know?

Expected Learning Outcome/s: **(LC Codes:** STEM_PC11AG-Ic-1-2**)**

1. Define an ellipse.
2. Graph an ellipse given an equation in standard form.

What's In?

Unlike circle and parabola, an ellipse is one of the conic sections that most students have not encountered formally before. Its shape is a bounded curve which looks like a flattened circle. The orbits of the planets in our solar system around the sun happen to be elliptical in shape (*see Figure 1).*

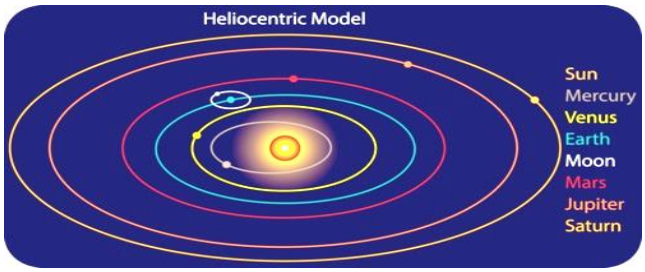
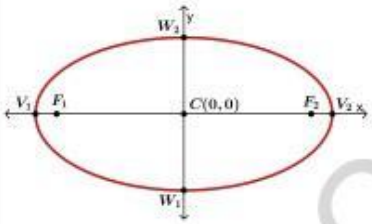
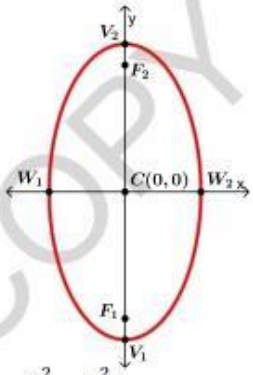
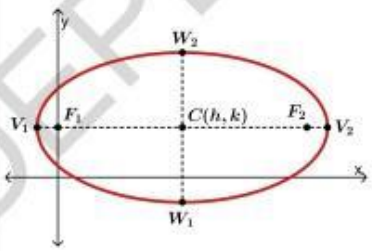
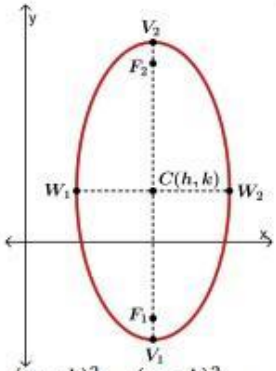


Figure 1 Source: *https://courses.lumenlearning.com/suny-earthscience/chapter/introduction-to-the-solar-system/*

Also, just like parabolas, ellipses have reflective properties that have been used in the construction of certain structures. These applications and more will be encountered in this lesson.

What's New?

An **ellipse** is the set all points in a plane the sum of whose distances from two fixed points in the plane is constant. The fixed points are called the **foci** (plural of focus) of the ellipse, and the line through them is sometimes called the **focal axis**. The point on the focal axis midway between the foci is the **center**, and the points where the ellipse crosses its focal axis are called the **vertices**. The line segment joining the two vertices is called the **major axis**; the line segment through the center, perpendicular to the major axis, and terminating at the ellipse is called the **minor axis**. The **eccentricity** of a curve tells us the amount of roundness of that curve. Now, the eccentricity of the circle is 0; that of the parabola is 1, while the eccentricity of the ellipse is between 0 and 1, that is $0 < e < 1$. A small eccentricity indicates that the ellipse tends toward being circular, whereas an eccentricity close to 1 indicates that the ellipse is elongated.

Center	Corresponding Graphs	
$(0,0)$	 $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1,\ a>b$	 $\frac{x^2}{b^2}+\frac{y^2}{a^2}=1,\ b>a$
(h,k)	 $\frac{(x-h)^2}{a^2}+\frac{(y-k)^2}{b^2}=1$ $a>b$	 $\frac{(x-h)^2}{b^2}+\frac{(y-k)^2}{a^2}=1$ $b>a$
	major axis: horizontal minor axis: vertical	major axis: vertical minor axis: horizontal

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NOTE : $c^2 = a^2 - b^2$	HORIZONTAL MAJOR AXIS	VERTICAL MAJOR AXIS
Standard Form	$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$	$\frac{(x-h)^2}{b^2} + \frac{(y-k)^2}{a^2} = 1$
Foci	F_1 (h – c, k)	F_1 (h, k - c)
	F_2 (h + c, k)	F_2 (h, k + c)
Vertices	V_1 (h – a, k)	V_1 (h, k - a)
	V_2 (h + a, k)	V_2 (h, k + a)
Co-vertices	W_1 (h, k - b)	W_1 (h – b, k)
	W_2 (h, k + b)	W_2 (h + b, k)

What's More?

Examples:

1. The foci of an ellipse are (−3, −6) and (−3, 2). For any point on the ellipse, the sum of its distances from the foci is 14. Find the standard equation of the ellipse.

   Solution. The midpoint (−3, −2) of the foci is the center of the ellipse. The ellipse is vertical (because the foci are vertically aligned) and c = 4. From the given sum, $2a = 14$ so $a = 7$. Also, $b = \sqrt{a^2 - c^2} = \sqrt{33}$ .The equation is

   $\frac{(x+3)^2}{33} + \frac{(y+2)^2}{49} = 1$.

2. Find an equation of the ellipse with center at (0,0), one focus at (3,0), and a vertex at (-4,0).

   Solution. Since its center at (0,0), the major axis coincide with the x-axis, one focus at (3,0), and vertex at (-4,0), then using the obtained equation,

$$a = -4\,; \quad c = 3$$
$$b^2 = a^2 - c^2$$
$$b^2 = (-4)^2 - (3)^2$$
$$b^2 = 16 - 9$$
$$b^2 = 7$$
$$b = 2.65$$

So, an equation of the ellipse is,

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$$

$$\frac{(x-0)^2}{(-4)^2} + \frac{(y-0)^2}{(2.65)^2} = 1$$

THE ELLIPSE

ELLIPSE - the set all points in a plane the sum of whose distances from two fixed points in the plane is constant.
FOCI - (plural of focus) fixed points of the ellipse.
FOCAL AXIS- line through the foci.
CENTER - The point on the focal axis midway between the foci.
VERTICES -the points where the ellipse crosses its focal axis.
MAJOR AXIS - The line segment joining the two vertices.
MINOR AXIS - line segment through the center, perpendicular to the major axis, and terminating at the ellipse.
ECCENTRICITY - tells us the amount of roundness of a curve.
ECCENTRICITY OF THE ELLIPSE - between 0 and 1, ($0 < e < 1$). Small eccentricity indicates that the ellipse tends toward being circular. If close to 1 indicates that the ellipse is elongated.

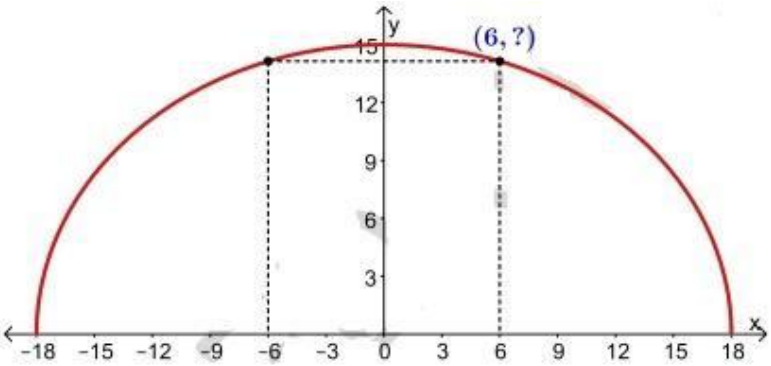
Summary:

NOTE : $c^2 = a^2 - b^2$	**HORIZONTAL MAJOR AXIS**	**VERTICAL MAJOR AXIS**
Standard Form	$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$	$\frac{(x-h)^2}{b^2} + \frac{(y-k)^2}{a^2} = 1$
Foci	F_1 (h – c, k)	F_1 (h, k - c)
	F_2 (h + c, k)	F_2 (h, k + c)
Vertices	V_1 (h – a, k)	V_1 (h, k - a)
	V_2 (h + a, k)	V_2(h, k + a)
Co-vertices	W_1 (h, k - b)	W_1 (h – b, k)
	W_2 (h, k + b)	W_2 (h + b, k)

What I Can Do?

Let us now apply the concept of ellipse to some situational problems.

A tunnel has the shape of a semi ellipse that is 15 ft high at the center, and 36 ft across at the base. At most how high should a passing truck be, if it is 12 ft wide, for it to be able to fit through the tunnel? Round off your answer to two decimal places.



Solution. Refer to the figure above. If we draw the semi ellipse on rectangular coordinate system, with its center at the origin, an equation of the ellipse which contains it, is

$$\frac{x^2}{18^2}+\frac{y^2}{15^2}=1$$

To maximize its height, the corners of the truck, as shown in the figure, would have to just touch the ellipse. Since the truck is 12 ft wide, let the point (6, n) be the corner of the truck in the first quadrant, where n>0, is the (maximum)height of the truck. Since this point is on the ellipse, it should fit the equation.

Thus, we have

$$\frac{6^2}{18^2}+\frac{n^2}{15^2}=1$$

$$n^2=15^2(1-\frac{6^2}{18^2})$$

Assessment

Give the coordinate of the center, vertices, covertices, and foci of the ellipse with the given equation.

1. $\frac{x^2}{169}+\frac{y^2}{25^2}=1$
2. $\frac{x^2}{144}+\frac{y^2}{169}=1$
3. $4x^2+13y^2=52$
4. $\frac{(x+7)^2}{16}+\frac{(y-2)^2}{25}=1$
5. $9x^2+16y^2+72x-96y+144=0$

What I Know?

Choose the letter of the best answer.

____**1**. What do you call to a conic section that is composed of pairs of parabolas whose openings are in opposite direction?

A. Circle B. Ellipse C. Hyperbola D. Parabola

____**2.** What do you call the line y = $\pm\frac{b}{a}$ in the standard form of equation of a hyperbola $\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$?

A. asymptotes B. vertices C. directrix D. foci

____**3**. What is $c^2 = a^2 + b^2$ in the standard form of equation of a hyperbola, $\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$?

A. asymptotes B. vertices C. directrix D. foci

For numbers 4-10. Please refer to this hyperbola equation $\frac{(x+1)^2}{16} - \frac{(y-3)^2}{48} = 1$

____**4.** What is the center of the hyperbola?

A. (1, -3) B. (-1, 3) C. (1, 3) D. (-1, 3)

____**5.** Which of the following is one of the vertices of the hyperbola?

A. (3, 3) B. (3, -3) C. (-3, 3) D. (-3, -3)

____**6.** Which of the following is one of the foci of the hyperbola?

A. (7, -3) B. (-7, -3) C. (-7, 3) D. (7, 3)

____**7.** What is the value of c ?

A. 4 units B. 8 units C. 12 units D. 14 units

____**8.** What is the length of conjugate axis? Round off the answer to nearest whole number.

A. 4 units B. 8 units C. 12 units D. 14 units

____**9.** What is the length of transverse axis?

A. 4 units B. 8 units C. 12 units D. 16 units

____**10.** Find the equation of the hyperbola.

A. $3x^2 - y^2 - 6x + 6y - 54 = 0$ C. $3x^2 - y^2 + 6x + 6y - 54 = 0$

B. $3x^2 + y^2 + 6x + 6y + 54 = 0$ D. $3x^2 - y^2 + 6x - 6y + 54 = 0$

Lesson 4 Hyperbolas

What I Need to Know?

Expected Learning Outcome/s: **(LC Codes:** STEM_PC11AG-Id-1-2**)**

1. Define a hyperbola.
2. Graph a hyperbola given an equation in standard form.

What's In?

The ***hyperbola*** is the locus of all points in a plane such that the absolute value of the difference of the distances from any point on the hyperbola to two given points in the plane, the *foci*, is constant.

$$| PF_1 - PF_2| = 2a$$

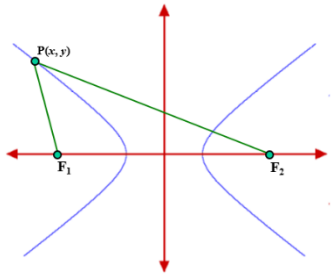


Figure 4.1

In Figure 4.1, given are the two points on the axis, $F_1(-c,0)$ and $F_2(c,0)$, the foci, both c units away from their midpoint (0,0). The midpoint is the center of the hyperbola.

What's New?

The features of the graph of a hyperbola with standard equation of $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.

Let c = $\sqrt{a^2 + b^2}$.

1. center: origin (0,0)
2. foci: $F_1(-c,0)$ and $F_2(c,0)$
3. vertices: $V_1(-a,0)$ and $V_2(a,0)$
   - vertices are points on the hyperbola and collinear with the center and foci

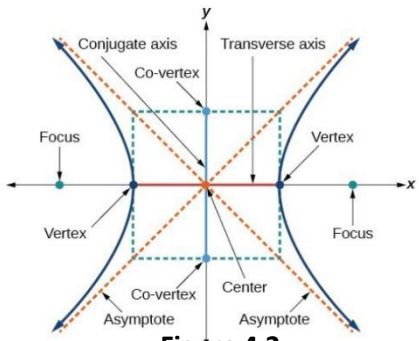


Figure 4.2

- V_1V_2 is called transverse axis and its length is 2a.

4. co – vertices: $W_1(0,- b)$ and $W_2(0,b)$
   - W_1W_2 is called conjugate axis and its length is 2b.
5. Asymptotes: $y = \frac{b}{a}x$ and $y = -\frac{b}{a}x$, the lines l_1 and l_2 in Figure 4.2
   - An auxiliary rectangle in Figure 4.2 has sides 2a and 2b with its diagonals intersecting at the center C.

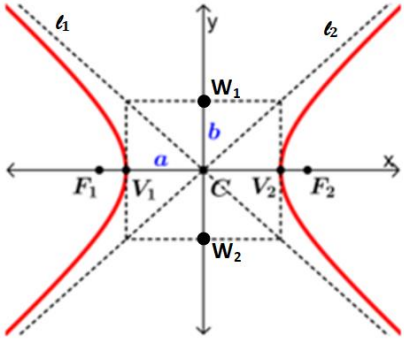


Figure 4.3

What is It?

Standard form of the hyperbola

An equation of the hyperbola centered at the origin:

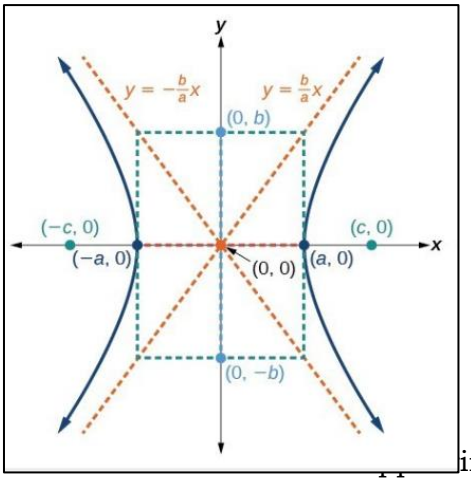


Figure 4.4

in the 1st term

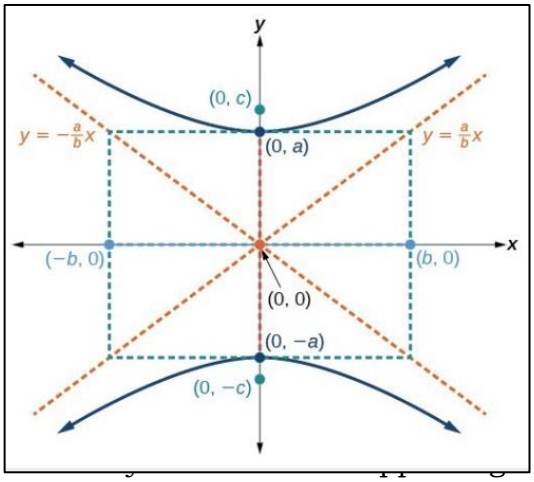


Figure 4.5

in the 1st term

An equation of the hyperbola centered at (h,k):

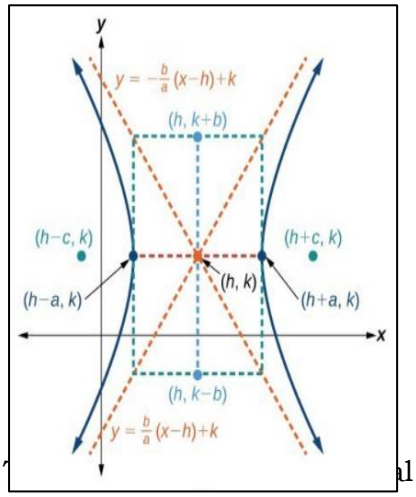


Figure 4.6

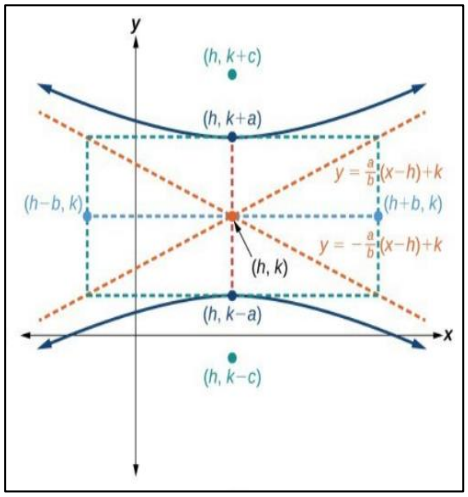


Figure 4.7

What's More?

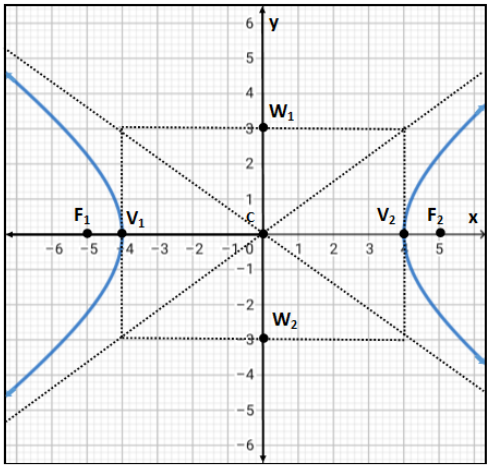
Analyzing the hyperbola

Given:	***Solution:***	***Reasons***
1. $\frac{x^2}{16} - \frac{y^2}{9} = 1$	$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$	formula
a. Find the center	a. C(0,0)	center at the origin
b. vertices	b. $(\pm 4,0)$ or (4,0) and (-4,0) $a^2 = 16$, a = 4	
c. co-vertices	c. $(0,\pm 3)$ or (0,3) and (0,-3) $b^2 = 9$, b = 3	
d. foci	d. $(\pm 5,0)$ or (5,0) and (-5,0) c = $\sqrt{a^2 + b^2}$. c = $\sqrt{16 + 9}$. c = $\sqrt{25}$ c = 5	
e. asymptotes	e. y = $\frac{b}{a}x$ and y = - $\frac{b}{a}x$ y = $\frac{3}{4}x$ and y = - $\frac{3}{4}x$	Formula
f. transverse axis		
g. conjugate axis		
h. graph		

f. 8 units $2a = 2(4) = 8$

g. 6 units $2b = 2(3) = 6$

h.



2. Write the standard equation of the hyperbola.

a. Having center at (2,- 3), a focus at (-3,-3) and a vertex at (6, −3).

Solution:

The center is (2,3), so h = 2 and k = - 3

The distance from center to focus is 5, c = 2 – (-3) = 5.

From the center and given vertex is 4, a = 6 – 2 = 4

Thus, b = $\sqrt{c^2 - a^2} = \sqrt{5^2 - 4^2} = \sqrt{9} = 3$.

Since the transverse axis is horizontal centered at (h,k),

$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$ is the formula.

The equation is

$$\frac{(x-2)^2}{16} - \frac{(y+3)^2}{9} = 1$$

b. $4x^2 - 9y^2 + 32x + 18y + 91 = 0$

Solution:

$$4x^2 + 32x - 9y^2 + 18y = -91$$

$$4(x^2 + 8x + ____) - 9(y^2 - 2y + ____) = -91$$

$$4(x^2 + 8x + 16) - 9(y^2 - 2y + 1) = -91 + 4(16) - 9(1)$$

$$4(x+4)^2 - 9(y-1)^2 = -36$$

$$\frac{4(x+4)^2}{-36} - \frac{9(y-1)^2}{-36} = \frac{-36}{-36}$$

$$\frac{-(x+4)^2}{9} - \frac{(y-1)^2}{4} = 1$$

$$\frac{(y-1)^2}{4} - \frac{(x+4)^2}{9} = 1$$

What I Have Learned?

- The ***hyperbola*** is the locus of all points in a plane such that the absolute value of the difference of the distances from any point on the hyperbola to two given points in the plane, the *foci*, is constant.

An equation of the hyperbola centered at the origin:

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \qquad \frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$$

An equation of the hyperbola centered at (h,k):

$$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1 \qquad \frac{(y-k)^2}{a^2} - \frac{(x-h)^2}{b^2} = 1$$

What I Can Do?

I. Given the equation of the hyperbola, determine the coordinates of the following:

2. $4x^2 - 3y^2 + 8x + 16 = 0$

   a. Find the center
   b. Vertices
   c. co- vertices
   d. foci
   e. Asymptotes
   f. transverse axis
   g. conjugate axis
   h. graph

II. Write the standard equation of the hyperbola.

A hyperbola has a focus at and vertices at .
$2x^2 - 3y^2 - 8x - 6y - 7 = 0$

Assessment

Short Answer. Supply the correct answer.

_____1. Is the set of all points (x,y) in a plane such that the difference of the distances between (x,y) and the foci is a positive constant.

_____2. It is the axis of a hyperbola that is perpendicular to the transverse axis and has the co-vertices as its endpoints.

_____3. It is the axis of a hyperbola that includes the foci and has the vertices as its endpoints.

_____4. What formula are we going to use if the center of the hyperbola is at (h,k) and the transverse axis is parallel to y-axis?

_____5. What formula are we going to use if the origin is the center of the hyperbola and the transverse axis is on x-axis?

_____6. Given the equation $\frac{x^2}{64} - \frac{y^2}{36} = 1$, what is the coordinates of the foci?

_____7. Given the equation $\frac{y^2}{16} - \frac{x^2}{9} = 1$, what is the coordinates of the vertices?

_____8. Given the equation $\frac{(x+1)^2}{16} - \frac{(y-3)^2}{9} = 1$, what is the length of the transverse axis?

_____9. Given the equation $\frac{(x+1)^2}{16} - \frac{(y-3)^2}{9} = 1$, what is the length of the conjugate axis?

_____10. Given the equation $\frac{(x+1)^2}{16} - \frac{(y-3)^2}{9} = 1$ What are the coordinates of the asymptotes?

Additional Activities

General Directions: Avoid erasure. Box your final answer.

I. Find the foci, vertices and asymptotes of the given hyperbola.

Equation of the hyperbola	*foci*	*vertices*	*asymptotes*
1. $4x^2 - y^2 = 256$			
2. $64y^2 - 9x^2 = 576$			

II. Write the standard form of the hyperbola.

Given	Standard form
1. Center at the origin; vertex at (3,0); focus at (6,0)	
2. Foci at (-6,3) and (4,3); vertices at (-5,3) and (3,3)	

Lesson 5 Identifying the Conic Section by Inspection

What I Need to Know?

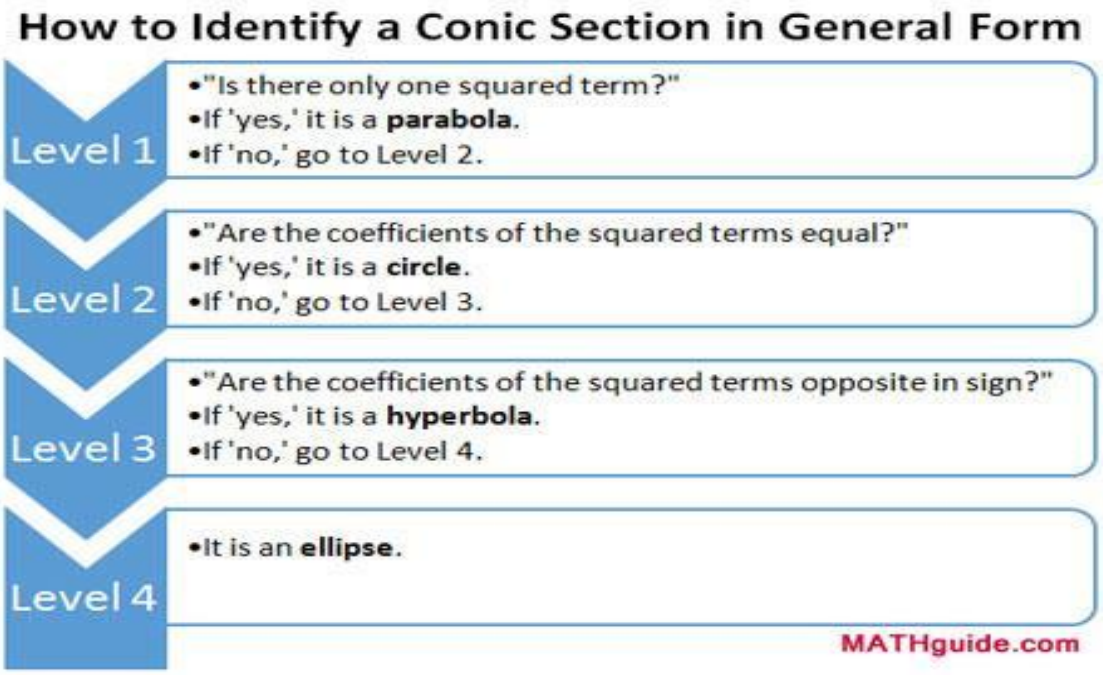
Expected Learning Outcome/s: **(LC Codes:** STEM_PC11AG-Ie-1**)**

1. Recognize the equation and important characteristics of the different types of conic sections.

What's In?

Let us recall the general form of the equations of the other conic sections. We may write the equations of conic sections we discussed in the general form $Ax^2 + By^2 + Cx + Dy + E = 0$.

To be able to identify these equations of conic sections in general form, we will make use of a graphic that will help us. The graphic below is called a process flow.



What's New?

It is only after transforming a given general equation to standard form that we can identify its graph either as one of the degenerate conic sections (*a point, two intersecting lines, or the empty set*) or as one of the non-degenerate conic sections (*circle, parabola, ellipse, or hyperbola*).

General Equation	Standard Equation	Graph
(A) $2x^2 + 2y^2 - 2x + 6y + 5 = 0$	$(x - \frac{1}{2})^2 + (y + \frac{3}{2})^2 = 0$	point
(B) $x^2 + y^2 - 6x - 8y + 50 = 0$	$(x - 3)^2 + (y - 4)^2 = -25$	empty set

In the given general equation, x^2 any y^2 are present then (A) and (B) are circle a non – degenerate conics. But after transforming it to standard equation, (A) is point and (B) is an empty set a degenerate conic sections. Thus, we can identify the given equation as one of degenerate or non- degenerate by transforming general equation to standard equation.

What is It?

Kinds of Conic Sections

- **Circle** - both x^2 and y^2 appear, and their coefficients are the same
  $Ax^2 + Ay^2 + Cx + Dy + E = 0$

  Example: $2x^2+2y^2-10x+12y-45 = 0$

  Degenerate cases: a point, and the empty set

- **Parabola** – exactly one of x^2 or y^2 appears
  $Ax^2 + Cx + Dy + E = 0$ ($D \neq 0$, opens upward or downward)
  $By^2 + Cx + Dy + E = 0$ ($D \neq 0$, opens to the right or left)

  Example: $x^2+4x - 5y+5=0$ opens upward
  $2y^2 -3x+4y-16 = 0$ opens to the right

- **Ellipse** - both x^2 and y^2 appear, and their coefficients A and B have the same sign and are unequal
  Examples: $3x^2+10y^2-10x+4y-10=0$
  Degenerate cases: a point, and the empty set

- **Hyperbola** - both x^2 and y^2 appear, and their coefficients A and B have different signs
  Examples: $3x^2-10y^2-10x+4y-10 = 0$

  Degenerate case: two intersecting lines

What's More?

The graphs of the following equations are non - degenerate conic sections. Identify the conic section.

1. $x^2 + y^2 - 2x - 4y - 4 = 0$
Answer: Circle
Reason: both x^2 and y^2 are present and same coefficient

2. $3x^2 + y + 12x + 1 = 0$
Answer: Parabola
Reason: x^2 is the only squared term and the graph opens downward

3. $9x^2 - 4y^2 - 36x - 40y - 388 = 0$
Answer: Hyperbola
Reason: both x^2 and y^2 appear and their coefficients 9 and -4 are different

4. $y^2 + 6x + 4y + 5 = 0$
Answer: Parabola
Reason: y^2 is the only squared term and the graph opens to the left

5. $x^2 + 9y^2 - 4x - 72y + 139 = 0$
Answer: Ellipse
Reason: both x^2 and y^2 appear, and their coefficients 1 and 9 are the same sign and unequal

What I Have Learned?

We may write the equations of conic sections we discussed in the general form ***$Ax^2 + By^2 + Cx + Dy + E = 0$***.

- ***Circle*** - both x^2 and y^2 appear, and their coefficients are the same
- ***Parabola*** – exactly one of x^2 or y^2 appears
- ***Ellipse*** - both x^2 and y^2 appear, and their coefficients A and B have the same sign and are unequal
- ***Hyperbola*** - both x^2 and y^2 appear, and their coefficients A and B have different signs

What I Can Do?

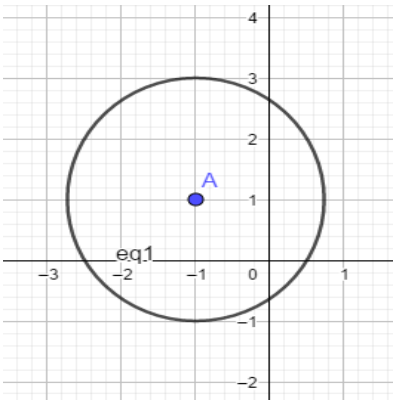
Given the following general equations, identify if it is a degenerate or non – degenerate conics and what are the specific graphs?

1. $4x^2 + 3y^2 + 8x - 6y = 5$

Possible answer: Ellipse, point, and empty set

Solution:

$$4x^2 + 3y^2 + 8x - 6y = 5$$
$$(4x^2 + 8x+) + (3y^2 - 6y+) = 5$$
$$4(x^2 + 2x+) + 3(y^2 - 2y+) = 5 + 4() + 3()$$
$$4(x^2 + 2x + 1) + 3(y^2 - 2y + 1) = 5 + 4(1) + 3(1)$$



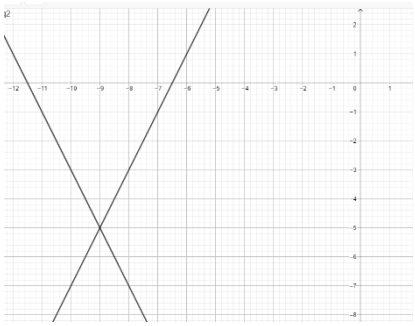
$$4(x+1)^2 + 3(y-1)^2 = 5 + 4 + 3$$
$$4(x+1)^2 + 3(y-1)^2 = 12$$
$$\frac{4(x+1)^2}{12} + \frac{3(y-1)^2}{12} = \frac{12}{12}$$
$$\frac{(x+1)^2}{3} + \frac{(y-1)^2}{4} = 1$$

Answer: *Non – Degenerate Conics, Ellipse*

2. $y^2 - 4x^2 - 72x + 10y - 299 = 0$
Possible answer: Hyperbola and two intersecting lines

Solution:

$$y^2 - 4x^2 - 72x + 10y - 299 = 0$$
$$y^2 + 10y - 4x^2 - 72x = 299$$
$$(y^2 + 10y+) - (4x^2 + 72x+) = 299$$
$$(y^2 + 10y+) - 4(x^2 + 18x+) = 299 + () - 4()$$
$$(y^2 + 10y+) - 4(x^2 + 18x+) = 299 + () - 4()$$
$$(y^2 + 10y + 25) - 4(x^2 + 18x + 81) = 299 + (25) - 4(81)$$
$$(y^2 + 10y + 25) - 4(x^2 + 18x + 81) = 299 + 25 - 324$$
$$(y+5)^2 - 4(x+9)^2 = 0$$
$$\frac{(y+5)^2}{4} - (x+9)^2 = 0$$



Answer: Degenerate conics, intersecting lines

Assessment

Directions: Answer the following questions about hyperbola.

_____1. What type of conic section is $x^2 + y^2 = 16$?

_____2. What type of conic section is $x = 4y^2$?

_____3. What conic section is the graph of $x = 2(y+3)^2 + 12$?

_____4. If the general form is $\boldsymbol{Ax^2 + By^2 + Cx + Dy + E = 0}$, how can you identify a circle in this form?

A. if A = B B. if A*B = 0 C. if A*B > 0, A≠B D. if A*B < 0

_____5. If the general form is $\boldsymbol{Ax^2 + By^2 + Cx + Dy + E = 0}$, how can you identify an ellipse in this form?

A. if A = B B. if A*B = 0 C. if A*B > 0, A≠B D. if A*B < 0

_____6. If the general form is $\boldsymbol{Ax^2 + By^2 + Cx + Dy + E = 0}$, how can you identify a parabola in this form?

A. if A = B B. if A*B = 0 C. if A*B > 0, A≠B D. if A*B < 0

_____7. If the general form is $\boldsymbol{Ax^2 + By^2 + Cx + Dy + E = 0}$, how can you identify a hyperbola in this form?

A. if A = B B. if A*B = 0 C. if A*B > 0, A≠B D. if A*B< 0

_____8. Which conic does not have a center?

_____9. Which of the following equation whose graph is a circle?

A. $2x^2+2y^2+3x+5y-10=0$

B. $x^2+y^2+4x-6y+13=0$

C. $x^2+y^2-18x+10y+106=0$

D. $20x^2+20y^2=0$

_____10. Which of the following equation whose graph is an intersecting lines?

A. $x^2-8x+15y+106=0$

B. $x^2+y^2+4x-6y+13=0$

C. $25y^2-25x^2-25=0$

D. $20x^2+20y^2=0$

Additional Activities

I. The graphs of the following equations are non - degenerate conic sections. Identify the conic section.

1. $3x^2+5y^2+6x-20y-13=0$
2. $16x^2-9y^2-96x+36y-36=0$

II. Given the following general equations, identify if it is a degenerate or non – degenerate conics and what are the specific graphs?

1. $25x^2-9y^2+50x-18y+16=0$
2. $x^2-4y^2-10x+24y-15=0$

Lesson 6 Problems Involving Different Conic Sections

What I Need to Know?

Expected Learning Outcome/s: **(LC Codes:** STEM_PC11AG-Ie-2**)**

1. Solves situational problems involving conic sections.

What's In?

Apollonius (262 B.C. – 200 B. C.) A Greek mathematician who made a thorough investigation of the conic sections. He considered himself a rival of Archimedes and was the first person to use the words *parabola, ellipse, and hyperbola.* His methods were expounded in an eight-volume work called *Conics,* and his work was so modern at that time and it is sometimes judged to be an *analytic geometry* preceding Rene Descartes. He believed that conic sections should be *study for the beauty of the demonstrations* rather than for practical applications.

What's New?

Conics, an abbreviation for conic sections, are curves that result from the intersection of a (right circular) cone and a plane. The intersection of the plane with the cone can take place either at the vertex of the cone or at any other part of the nappe either below or above the vertex.

What is It?

In solving real life problems involving conic section it is require to use the properties of the different type of conics like the property of circle, parabola, ellipse and hyperbola.

What I Have Learned?

The Parabola (Parabola with Vertex at the Origin, and Parabola with Vertex at (*h,k)); The Ellipse* (Ellipse with Center at the Origin, and Ellipse with center at *(h,k);* and the Hyperbola (Hyperbola with Center at the Origin, and hyperbola with Center at *(h,k))*

What I Can Do?

Sample Problems:

1. A circle has center at the focus of the parabola $y^2 + 16x + 4y = 44$. and is tangent to the directrix of this parabola. Find its standard equation.

   Solution. The standard equation of the parabola is $(y+2)^2 = -16(x-3)$. Its vertex is V (3; −2). Since 4c = 16 or c = 4, its focus is *F (−1; −2)* and its directrix is x = 7. The circle has center at (−1; −2) and radius 8, which is the distance from F to the directrix. Thus, the equation of the circle is $(x+1)^2 = (y+2)^2 = 64$

2. The vertices and foci of $5x^2 - 4y^2 + 50x + 16y + 29 = 0$ are, respectively, the foci and vertices of an ellipse. Find the standard equation of this ellipse.

   Solution. We first write the equation of the hyperbola in standard form:

$$\frac{(x+5)^2}{16} - \frac{(y-2)^2}{20} = 1.$$

For this hyperbola, using the notations a_h, b_h, and c_h to refer to a, b, and c of the standard equation of the hyperbola, respectively, we have $a_h = 4$, $b_h = 2\sqrt{5}$, $c_h = \sqrt{a_h^2 + b_h^2} = 6$, so we have the following points:

center: (−5; 2)
vertices: (−9; 2) and (−1; 2)
foci: (−11; 2) and (1; 2).

It means that, for the ellipse, we have these points

center: (−5; 2)
vertices: (−11; 2) and (1; 2)
foci: (−9; 2) and (−1; 2)

In this case, $c_e = 4$ and $a_e = 6$, so that $b_e = \sqrt{a_e^2 + b_e^2} = \sqrt{20}$. The standard equation of the ellipse is

$$\frac{(x+5)^2}{36} + \frac{(y-2)^2}{20} = 1.$$

Assessment

Problem Solving.

1. Find the standard equation of the hyperbola one branch of which has focus and vertex that are the same as those of $x^2 - 6x + 8y = 23$, and whose conjugate axis is on the directrix of the same parabola.
2. An ellipse has equation $25x^2 + 16y^2 + 150x - 32y = 159$. Find the standard equations of all parabolas whose vertex is a focus of this ellipse and whose focus is a vertex of this ellipse.
3. Find the standard equation of the hyperbola whose conjugate axis is on the directrix of the parabola $y^2 + 12x + 6y = 39$, having the focus of the parabola as one of its foci, and the vertex of the parabola as one of its vertices.
4. Find the standard equation of the parabola opening to the left whose axis contains the major axis of the ellipse $x^2 + 4y^2 - 10x - 24y + 45 = 0$, whose focus is the center of the ellipse, and which passes through the covertices of this ellipse.
5. Find the standard equation of the ellipse whose major and minor axes are the transverse and conjugate axes (not necessarily in that order) of the hyperbola $4x^2 - 9y^2 - 16x - 54y = 29$.

Additional Activities

1. Find the standard form of the equation of the circle whose center is at (2,1) and whose radius is 2.
2. Find the equation of parabola with vertex at (-1, -4), and with y= 5 as the line of directrix.
3. Find an equation of the ellipse with center at (0, 0), one focus at (3,0), and a vertex at (-4,0),

Lesson 7 Review of Sequence and Series

What I Need to Know?

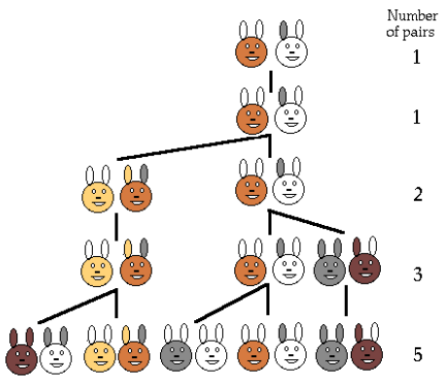
Expected Learning Outcome/s: **(LC Codes:** STEM_PC11SMI-Ih-1-2**)**

1. Illustrate a series.
2. Differentiate a series from a sequence.

What's In?

Suppose a newly-born pair of rabbits, one male, one female, are put in a field. Rabbits are able to mate at the age of one month so that at the end of its second month a female can produce another pair of rabbits. Suppose that our rabbits never die and that the female always produces one new pair (one male, one female) every month from the second month on. How many pairs will there be in one year?

Solution: They were fully grown after one month and did what rabbits do best, so that the next month two more baby rabbits (again a boy and a girl) were born.



Eventually, the number of pairs of rabbits in the field are 1, 1, 2, 3, 5, 8, 13, 21, 34, ... Can you see how the series is formed and how it continues? What pattern do you see?

In the Fibonacci Sequence, each term is the sum of the two proceeding terms. For instance, 1 + 1 = 2, 2 + 3 = 5, 3 + 5 = 8. The first 12 numbers in the Fibonacci Sequence are 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89 and 144.

So how many pairs will there be in one year?

Answer: In one year, there will be 144 pairs of rabbits.

In everyday language, the term **"sequence"** means a succession of things in a definite order – chronological order, size order, or logical order. In mathematical, the term **"sequence"** is commonly used to denote a succession of numbers whose order is determined by rule or function. In this lesson, we will review the definition and types of sequences and series.

What's New?

- **Sequence** - is a function whose domain is the set of positive integers or the set {1, 2, 3,...,n}.
- Example: 2,4,6,8,10,...
- **Term** – the individual elements if a sequence.
- **Finite Sequence** – a sequence with a last term
- Example: 4,9,16,25,36
- **Infinite Sequence** – a sequence which do not have a last term. It has ellipsis (...) indicating that the configuration continues without end.
- Example: 16,21,16,31,36,...
- **Series** – it represents the sum of the terms of a sequence.
- a sum of numbers (separated by "+" or " - " sign)
- Example: 2+4+6+8+10
- **Arithmetic sequence** - a sequence in which each term after the first is obtained by adding a constant (called the common difference) to the preceding term.
- Example: 6,9,12,15,18 common difference or d = 3
- **Geometric sequence** - a sequence in which each term after the first is obtained by multiplying the preceding term by a constant (called the common ratio).
- Example: 4,8,16,32,64 common ration or r = 2

What is It?

A *sequence* of numbers is an ordered collection of numbers t_1, t_2,...,t_n,... in which the first number t_1, is called the first term, the second number t_2, is called the second term, and so forth, and t_n, the nth term. The dots are used to indicate that the sequence continues indefinitely.

A *series* represents the sum of the terms of a sequence. It is separated by "+" or " - " sign.

The sequence with nth term t_n is usually denoted by $\{t_n\}$ and the associated series is given by $S = t_1 + t_2 + t_3 + \ldots + t_n$.

Example: Determine the first five terms of each defined sequence, and give their associated series.

1. $a_n = n^3$
2. $a_n = 2n - 1$
3. $a_n = \frac{1}{2^n}$

Solution. We denote the nth term of a sequence by a_n, and
$S = a_1 + a_2 + a_3 + a_4 + a_5$.

1. $a_n = n^3$

First five terms: $a_1 = (1)^3 = 1$, $a_2 = (2)^3 = 8$, $a_3 = (3)^3 = 27$, $a_4 = (4)^3 = 64$, $a_5 = (5)^3 = 125$

Associated series: S = $a_1 + a_2 + a_3 + a_4 + a_5$ = 1+8+27+64+125 = 225

2. a_n = 2n – 1
First five terms: a_1 = 2(1) – 1 = 1, a_2 = 2(2) – 1 = 3, a_3 = 2(3) – 1 = 5,
a_4 = 2(4) – 1 = 7, a_5 = 2(5) – 1 = 9
Associated series: S = 1+3+5+7+9 = 25
3. $a_n = \frac{1}{2^n}$
First five terms: $a_1 = \frac{1}{2^1} = \frac{1}{2}$, $a_2 = \frac{1}{2^2} = \frac{1}{4}$, $a_3 = \frac{1}{2^3} = \frac{1}{8}$, $a_4 = \frac{1}{2^4} = \frac{1}{16}$,
$a_5 = \frac{1}{2^5} = \frac{1}{32}$
Associated series: S = $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{32} = \frac{31}{32}$

What's More?

An ***arithmetic sequence*** is a sequence in which each term after the first is obtained by adding a constant (called the common difference) to the preceding term.

If the **nth** term of an arithmetic sequence is a_n and the common difference is d, then $a_n = a_1 + (n-1)\,d$.

The associated arithmetic series with n terms is given by $S_n = \frac{n\,(a_1+a_n)}{2} = \frac{n\,[2a_1+(n-1)d]}{2}$

Example: Find the 32nd term in the arithmetic sequence 23,32,41,...

Solution: a_1= 23 (1st term), n = 32 (number of term),
d = 9 (common difference), a_n = ? (Last term)
$a_n = a_1 + (n-1)\,d$ (Formula)
$a_{32} = 23 + (32-1)\,9$
$a_{32} = 23 + (31)\,9$
$a_{32} = 23 + 279$
$a_{32} = 302$
The 32nd term in the sequence is therefore 302.

Associated arithmetic series: $S_{32} = \frac{32\,(23+302)}{2}$ = **5,200**

A ***geometric sequence*** is a sequence in which each term after the first is obtained by multiplying the preceding term by a constant (called the common ratio).

If the nth term of a geometric sequence is a_n and the common ratio is r, then

$$a_n = a_1 r^{n-1}$$

The associated geometric series with n terms is given by

$$S_n = \begin{cases} na_1 & , \text{ if } r=1 \\ \frac{a_1(1-r^n)}{1-r} & , \text{if } r \neq 1 \end{cases}$$

Example: Find the 10th term in the geometric sequence -15, 45, -135, ...

Solution: a_1= -15 (1st term), n = 10 (number of term),

r = -3 (common ratio), a_n = ? (Last term)
$a_n = a_1 r^{n-1}$ (Formula)
$a_{10} = (-15)(-3)^{10-1}$
$a_{10} = (-15)(-3)^9$
$a_{10} = (-15)(-19683)$
$a_{10} = 295{,}245$

The 10th term in the sequence is therefore 295,245.

Associated geometric series:

Since r = -3, the formula is $S_n = \frac{a_1(1-r^n)}{1-r}$

$$S_{10} = \frac{-15[1-(-3)^{10}]}{1-(-3)} = \frac{-15[1-(59049)]}{1+3} = \frac{-15(-59048)}{4} = \frac{885720}{4} = \mathbf{221{,}430}$$

What I Have Learned?

Answer the following questions:

1. What is a sequence?
2. What is a series?
3. What is an arithmetic sequence?
4. What is a geometric sequence?
5. What is the difference between arithmetic and geometric sequence?

What I Can Do?

I. Discover the pattern in each set of numbers below then find the next two numbers.

1. 2, 9, 16, 23, 30, _____, _____ 37, 44
2. -8, -16, -32, -64, -128, ____, _____, -256, -512
3. 54, 39, 24, 9, -6, ____, ____, -21, -36
4. $\frac{2}{3}, \frac{11}{12}, 1\frac{1}{6}, 1\frac{5}{12}, 1\frac{2}{3}$, _____, _____, $1\frac{11}{12}, 2\frac{1}{6}$
5. 9, 16, 25, 36, ____, _____, 49, 64

II. Determine the first five terms of each defined sequence and give their associated series.

1. $a_n = 7n - 3$
2. $a_n = n^3 - 5$
3. $a_n = (-3)^n$
4. $a_n = 4^{n+2}$
5. $a_n = \frac{2n+1}{n+2}$

III. Determine whether the following sequences are Arithmetic, Geometric or Fibonacci sequence.

1. 67, 54, 41, 28, 15, ...
2. 3, 6, 9, 15, 24, ...
3. 2, 2, 2, 2, 2, ...
4. 3, -12, 48, -192, 768, ...
5. $1, \frac{5}{3}, \frac{7}{3}, 3, \frac{11}{3}, \ldots$

Assessment

Short Answer. Supply the correct answer.

1. Which of the following is an infinite arithmetic sequence?
   a. 4,5,6,7,...,11 c. 7,11, 15, ..., 27
   b. 5,7,9,11,...,19 d. 9, 13, 17, 21, ...
2. What is constant in arithmetic sequence?
   a. common difference c. sum
   b. common ratio d. notation
3. What is constant in a geometric sequence?
   a. common difference c. sum
   b. common ratio d. notation
4. Which of the following is an arithmetic sequence?
   a. 5, 15, 45, ... c. -7, -5, -3, -1, 1, ...
   b. 1, 2, 3, 5, 8, ... d. 6, -12, 24, -48, ...
5. Which of the following is the sum of the first 12 terms of the sequence 48, 43, 38, 33, 28, ...?
   a. 230 c. 244
   b. 234 d. 246

Answer the following problems:

1. A review center opened with only 8 students enrolling on the first month. However, it has been observed that the number of students doubles every month. How many students were enrolled on the 10th month?

2. Find the sum of all odd integers from 1 to 100.

Additional Activities

Find the 10th term and the sum of the first 10 terms in a geometric sequence if the 1st term is 8 and the 4th term is -2

Lesson 8 Sigma Notation

What I Need to Know?

Expected Learning Outcome/s: **(LC Codes:** STEM_PC11SMI-Ih-3**)**

1. Use sigma notation to represent a series.
2. Apply the use of sigma notation in finding the sum.

What's In?

The sigma notation is a shorthand for writing sums. In this lesson, we will see the power of this notation in computing sums of numbers as well as algebraic expressions. You may use this link for checking of your answers, https://goodcalculators.com/summation-calculator/

What's New?

We now introduce a special notation for writing sum of the terms of a sequence. This notation is called the **sigma Notation** also known as **summation notation.** It involves the use of the capital Greek letter for "S", called ***sigma***, which is denoted by **Σ**. We will see the use of the symbol ***i*** which is called the **index of the summation**. The index serves as dummy variable because it can be replaced by any other symbols. If ***f(i)*** denotes some functional expression involving variable ***i***, then *f(m) + f(m + 1) + f(m + 2) + · · · + f(n) can be compac*tly written in sigma notation, and we write it as

$$\sum_{i=m}^{n} f(i)$$

which is read "the summation of ***f(i)*** from $i = m \text{ to } n$. " Where, ***m*** and ***n*** are integers with $m \leq n, f(i)$ is a term of the summation, and the letter ***i*** is the index, **m** the **lower bound** and **n** is the **upper bound**. The value of ***i*** starts at ***m***, and ends at **n**.

What's More?

Expand each summation and simplify if possible.

$$1.)\sum_{i=1}^{5} 2i = 2(1) + 2(2) + 2(3) + 2(4) + 2(5) = 2 + 4 + 6 + 8 + 10 = 30$$

$$2.)\sum_{i=0}^{5} 2^i = 2^0 + 2^1 + 2^2 + 2^3 + 2^4 + 2^5 = 1 + 2 + 4 + 8 + 16 + 32 = 63$$

$$3.)\sum_{i=2}^{4} (2i + 3) = [(2(2) + 3] + [2(3) + 3] + [2(4) + 3] = 7 + 9 + 11 = 27$$

Associated with sequence $a_1, a_2, a_3 \ldots, a_n$ is the series

$$a_1, a_2, a_3 \ldots, a_n = \sum_{i=1}^{n} a_1$$

Specifically, for the sequence 4, 8, 12, 16, 20,24, the associated series is

$$4 + 8 + 12 + 16 + 20 + 24 = \sum_{i=1}^{6} 4i$$

Writing series using the sigma notation.

3+6+9+12+15

Solution:

We have already established that the sequence whose terms are given above has the general term $a_n = 3n$. The terms 3,6,9,12,15 are obtained by replacing n with 1,2,3,4,5, respectively. Hence,

$$3 + 6 + 9 + 12 + 15 = \sum_{i=1}^{5} 3i$$

What I Have Learned?

The **sigma notation** is defined by the equation

$$\sum_{i=m}^{n} f(i) = f_m + f_{m+1} + f_{m+2} + \cdots + f_{n-1+}f_n$$

What I Can Do?

Where, ***m*** and ***n*** are integers with $m \leq n, f(i)$ is a term of the summation, and the letter ***i*** is the index, **m** the **lower bound** and **n** is the **upper bound**. The value of ***i*** starts at ***m***, and ends at **n**.

Write the sigma notation for the following series. Use of the symbol *i* for the index of the summation.

1. 7+14+21+28+35
2. 6+3+0+(-3) + (-6) + (-9)
3. $2^3+2^5+2^7+2^9+2^{11}+2^{13}$

Expand each summation and simplify if possible.

4.) $\sum_{i=1}^{7} (4i+3)$

5.) $\sum_{i=1}^{5} -$

Assessment

Rewrite each series as a sum.

1.) $\sum_{i=0}^{5} i$

2.) $\sum_{m=1}^{6} (4m^2 + 3)$

3.) $\sum_{a=1}^{6} (3a^2 - 1)$

4.) $\sum_{k=4}^{6} (4k^2 + 3)$

5.) $\sum_{i=1}^{5} (500 - i)$

Rewrite each series using sigma notation. Use of the symbol *i* for the index of the summation.

6.) 1 + 2 + 3 + 4

7.) 3 + 9 + 27 + 81

8.) 1 + 4 + 9 + 16 + 25

9.) 4 + 8 + 12 + 16

10.) $\frac{1}{2} + \frac{2}{3} + \frac{3}{4} + \frac{4}{5} + \frac{5}{6}$

Additional Activities

Write the sigma notation for the following series.

1.) 1+ (-3) +5+(-7) + 9 + (-11) +… (-29)

Expand each summation and simplify if possible.

2) $\sum_{i=1}^{7} 3i^2(2i - 1) =$

References

- Pre-Calculus for Senior High School, Anthony Cueno et al
- Pre-Calculus, John Gabriel P. Pelias
- Teaching Guide for Senior High School: Pre-Calculus
- Comandante, F.L. (2009). *Analytic geometry: Made easy*. Mandaluyong City: National Book Store.
- Garces, I.J.L., et.al. (2016). *Precalculus: Teaching guide for senior high school*. Quezon City: Commission on Higher Education
- Villaluna, T. (2017). *Pre-calculus*: *Analysis and applications*. Valenzuela City: Tru-Copy Publishing House, Inc.
- https://www.khanacademy.org/math/geometry/hs-geo-circles/hs-geo-circle-standard-equation/v/radius-and-center-for-a-circle-equation-in-standard-form
- https://courses.lumenlearning.com/waymakercollegealgebra/chapter/equations-of-hyperbolas/
- https://www.khanacademy.org/math/precalculus/x9e81a4f98389efdf:conics/x9e81a4f98389efdf:hyperb-intro/v/conic-sections-intro-to-hyperbolas
    - https://www.khanacademy.org/math/precalculus/x9e81a4f98389efdf:conics/x9e81a4f98389efdf:non-origin-hyperb/v/conic-sections-hyperbolas-3
- http://www.mathguide.com/lessons/ConicSections.html
- https://www.khanacademy.org/math/geometry/hs-geo-circles/hs-geo-circle-standard-equation/v/radius-and-center-for-a-circle-equation-in-standard-form
- https://www.youtube.com/watch?v=0TReLDtuafo
- EC – Math by Vista, Fernandez, Pulido and Enojado
- https://educ.queensu.ca/sites/webpublish.queensu.ca.educwww/files/files/Community/COC/Melodies/Mister%20Fibonacci%20-%20Notes.pdf

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